ENGINEERING GRAPHICS • DIGITAL NOTES

# UNIT IV

## Sections of Solids & Development of Surfaces

*Cut it open, then unfold it flat.*

**SYLLABUS COVERED**

- ▸ Perpendicular & inclined section planes
- ▸ Sectional views & true shape of section
- ▸ Sections of solids in simple positions
- ▸ Parallel-line & radial-line development
- ▸ Development: cube, prism, cylinder, pyramid, cone

## 1. Sections of Solids – Basics

**DEFINITION**

**Section plane (cutting plane)** – an imaginary plane that cuts the solid. **Section** – the figure formed on the cut surface. **Sectional view** – the view of the remaining solid after the portion on the observer's side is removed. **True shape of section** – the actual shape of the cut surface, seen when looking ⊥ to the cutting plane.

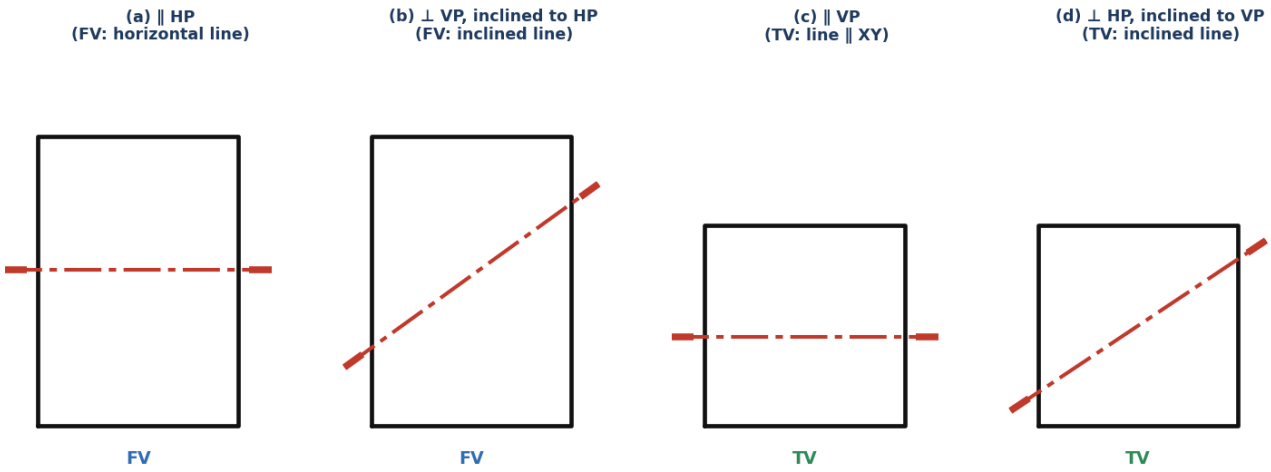



*Fig. 1.1 — Common positions of cutting planes (red chain line = trace of the cutting plane)*

| Section plane | Seen as a line in | True shape of section seen in |
|---|---|---|
| ‖ HP (⊥ VP) | FV (horizontal line) | **TV** – directly |
| ‖ VP (⊥ HP) | TV (line ‖ XY) | **FV** – directly |
| ⊥ VP, inclined θ to HP (AIP) | FV (inclined line) | Auxiliary view on a plane ‖ cutting plane |
| ⊥ HP, inclined φ to VP (AVP) | TV (inclined line) | Auxiliary view on a plane ‖ cutting plane |

**PROCEDURE**

1. Draw FV and TV of the **solid in simple position**. Draw the **cutting-plane trace** in the view where it appears as a line (chain line, thick at ends).
2. Mark where the trace meets each **edge / generator** in that view: points **1, 2, 3 …** Name them on the trace.
3. Project these points to the other view (verticals / horizontals) on the corresponding edges to get the **section points**; join in order.
4. **Remove** the portion between the observer and the plane; draw the remaining outline. **Hatch** the section at 45°.
5. For the **true shape**: draw $X_1Y_1$ ‖ trace, project ⊥ to the trace from each point and make distances = those in the other view from XY.

## 2. Sections of Prism, Pyramid, Cylinder & Cone

### Square prism (40 × 70) cut by a plane ⊥ VP inclined 40° to HP

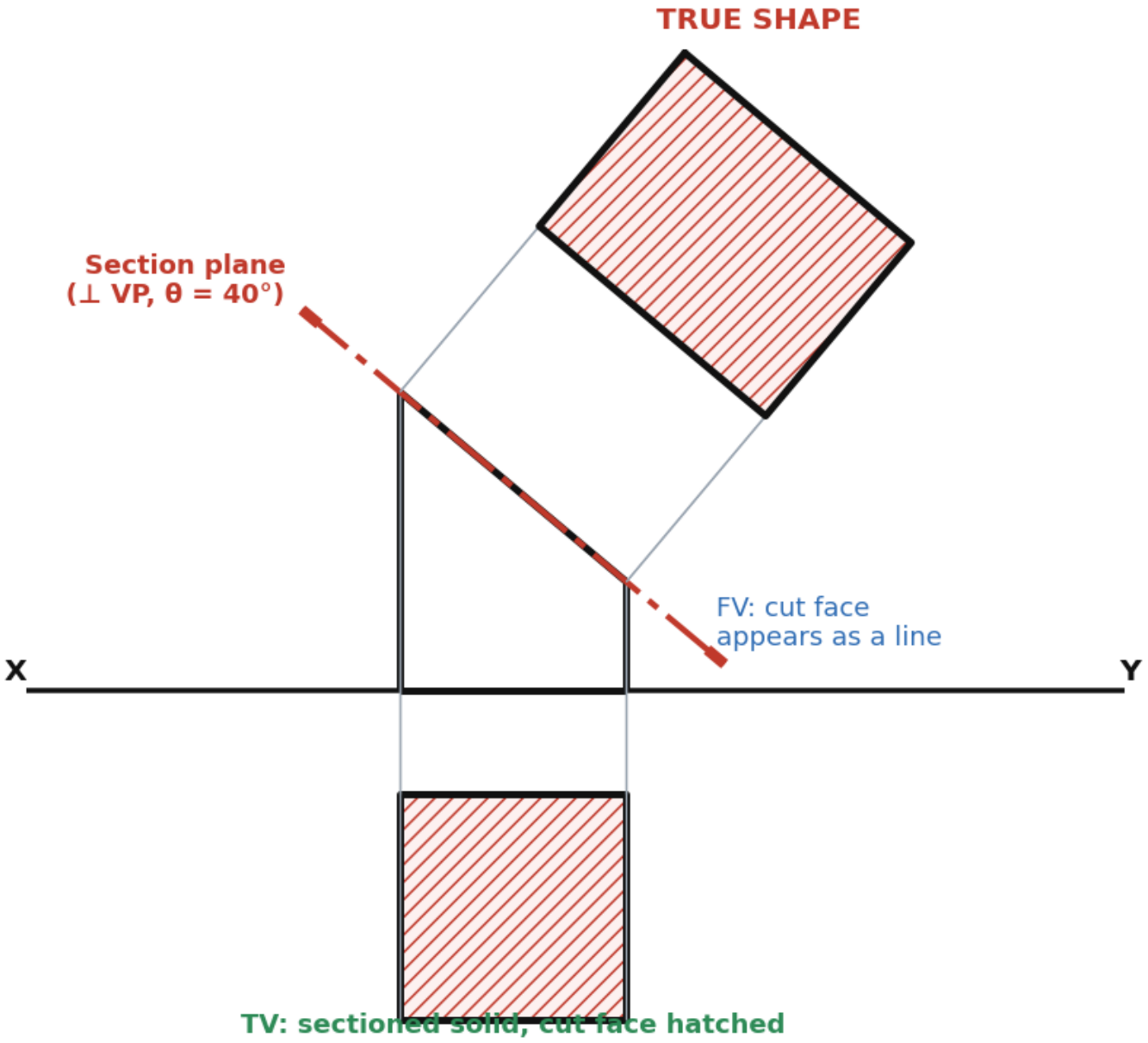



*Fig. 2.1 — Square prism cut by an inclined plane (AIP): FV, TV (hatched) and true shape*

**KEY POINTS**

- A plane ⊥ VP and inclined to HP cuts a **prism** in a figure of the **same number of sides as the base** (here a rectangle).
- The TV of the section lies **inside the base outline**; its true length is the **sloping length of the cut line** in the FV (longer than its TV length), while its width equals the base width.
- Distances for the true shape are taken from the **TV** (distance from XY) – the FV only supplies the direction.

## Square pyramid (40 × 70) cut by a plane ⊥ VP inclined 35° to HP

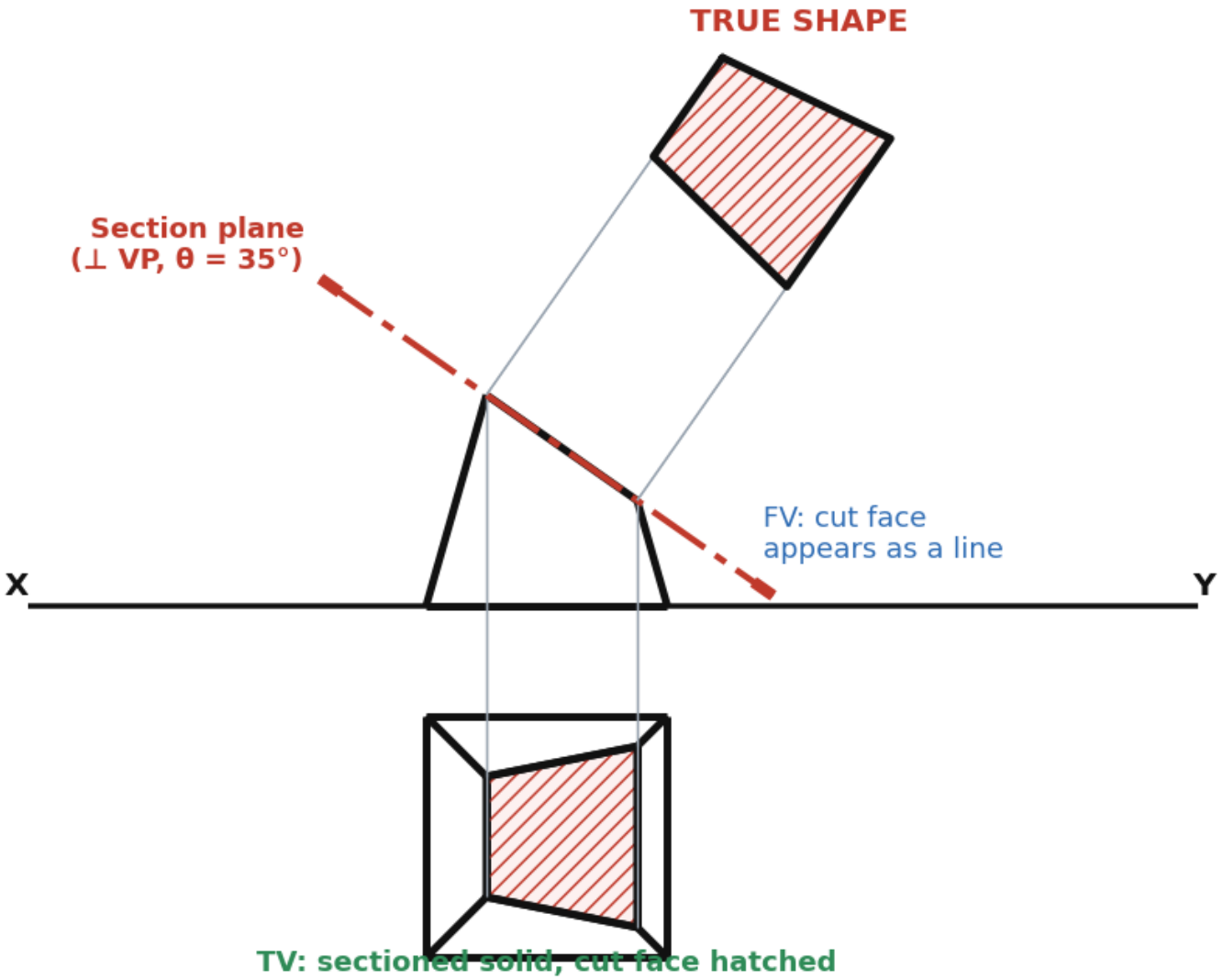



*Fig. 2.2 — Square pyramid cut by an inclined plane*

For the **pyramid**, each cut point lies on a slant edge; its position in the TV is found by **projecting the FV point to the corresponding edge** (apex-to-corner line). If the plane cuts the base, include the points where it meets the base edges.

## Cone (r = 25, h = 65) cut by a plane ⊥ VP inclined 38° to HP

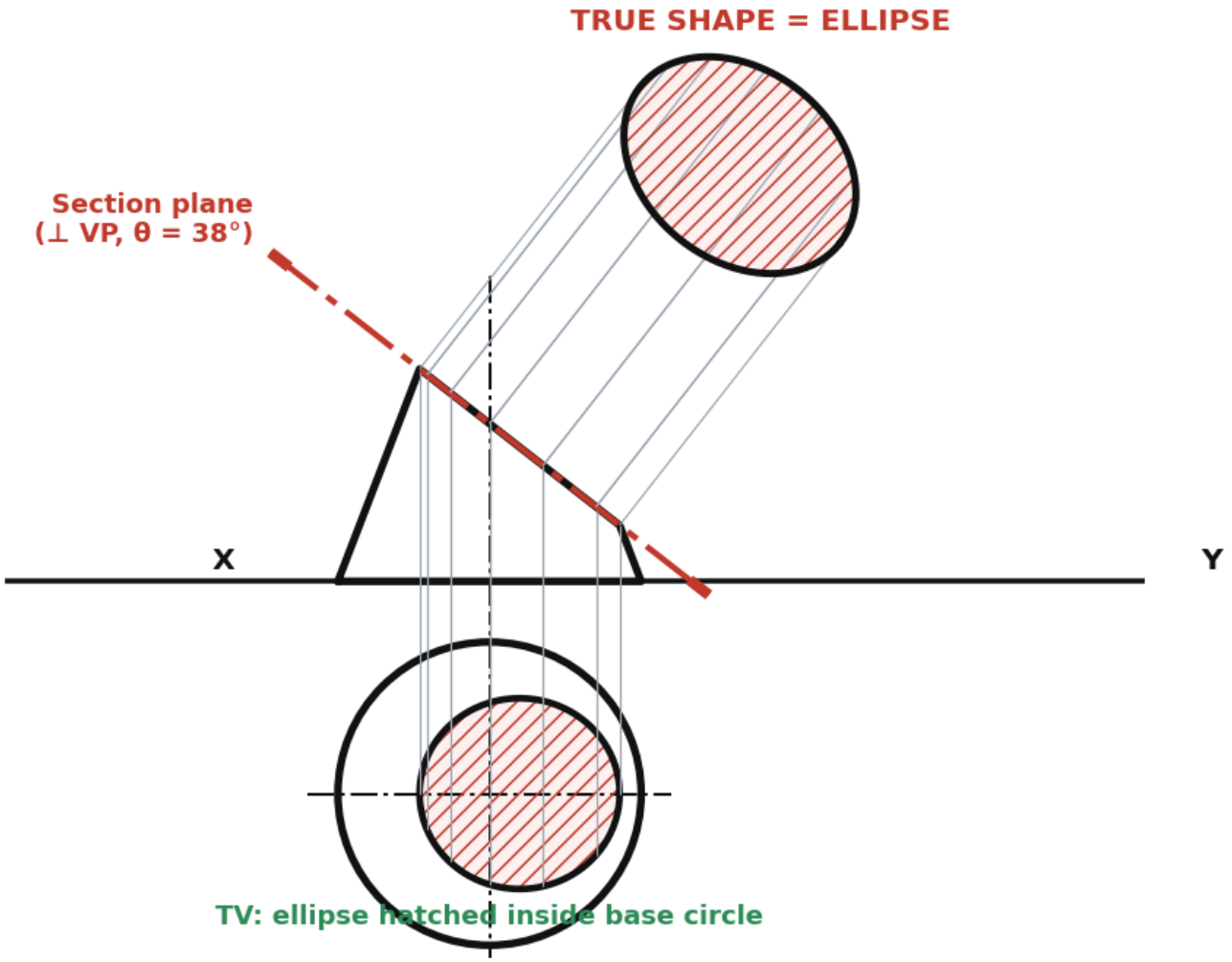



*Fig. 2.3 — Cone cut by an inclined plane – the true shape is an ellipse*

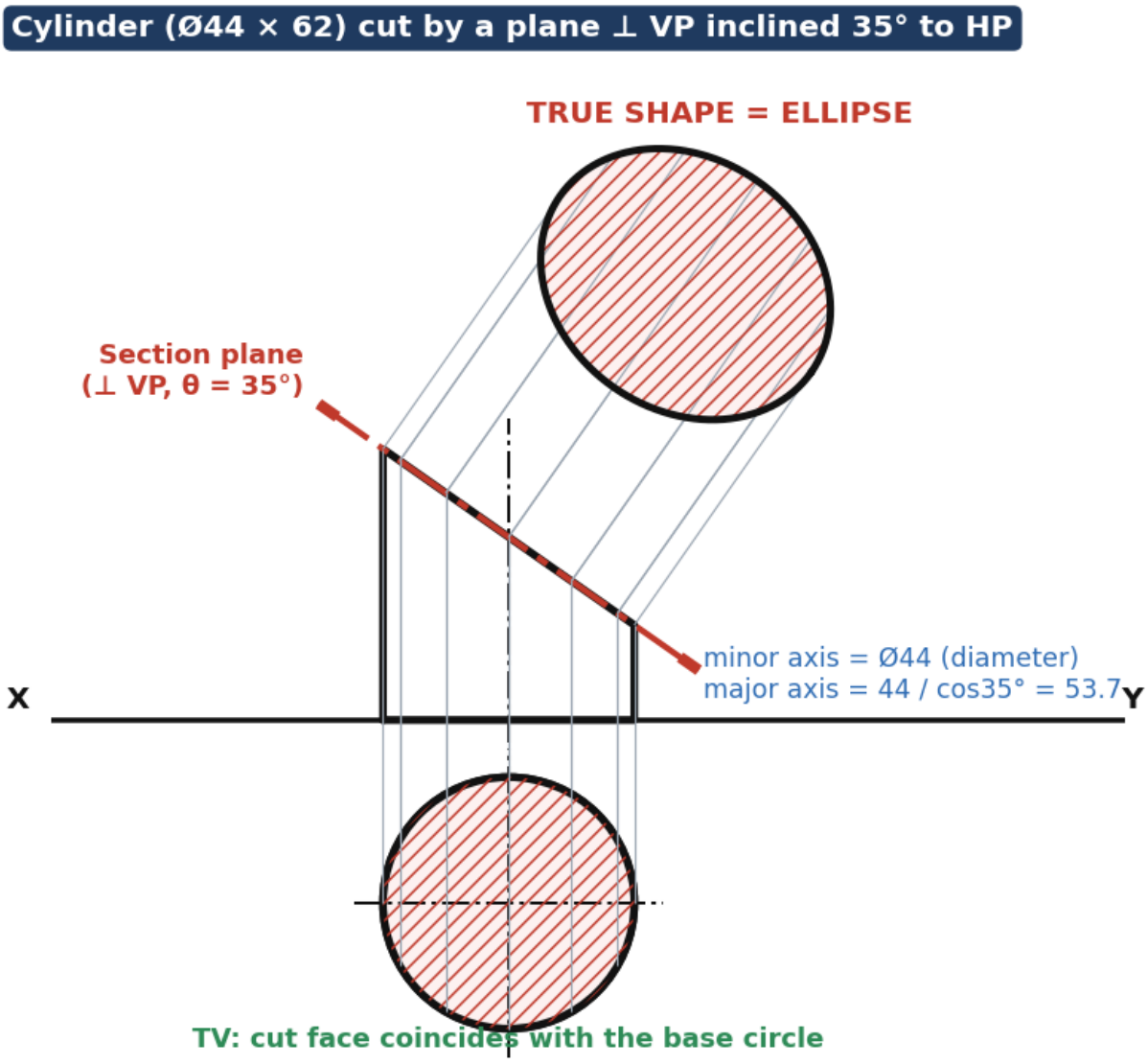



*Fig. 2.4 — Cylinder cut by an inclined plane*

| Cutting plane on a cone | Section | Condition |
|---|---|---|
| ⊥ axis | Circle | Parallel to base |
| Inclined, cuts all generators | Ellipse | θ is **less steep** than the generators (θ < angle of generator with base) |
| ∥ one generator | Parabola | θ = angle of generator |
| Steeper than generators (incl. ∥ axis) | Hyperbola | θ > angle of generator |
| Through the apex | Triangle | Contains the apex |

**EXAM TIP**

**Cylinder:** plane inclined to the axis cuts an **ellipse**; plane ∥ axis cuts a **rectangle**; plane ⊥ axis cuts a **circle**. Divide the base circle into 12 equal parts and draw 12 generators – project each cut point to get the ellipse.

**SOLVED EXAMPLE**

**A square pyramid (base 40, axis 70) is cut by a plane ⊥ VP inclined 35° to HP through a point 28 mm up the axis. Find the section.**

Cutting line in FV: $z = 28 - (x - x_{axis})\cdot\tan35°$. It meets the left and right slant edges at heights **35 mm** and **17.5 mm**; the length of the cut line in the FV (= true length of the section) is $\sqrt{(25^2 + 17.5^2)} \approx$ **30.5 mm**. Project the points to the TV on the same edges → quadrilateral section; draw the true shape in the auxiliary view, hatch it, and complete the remaining solid.

## 3. Sectional Views & Hatching

**Sectional view: solid cut by plane A–A, the cut metal is hatched**

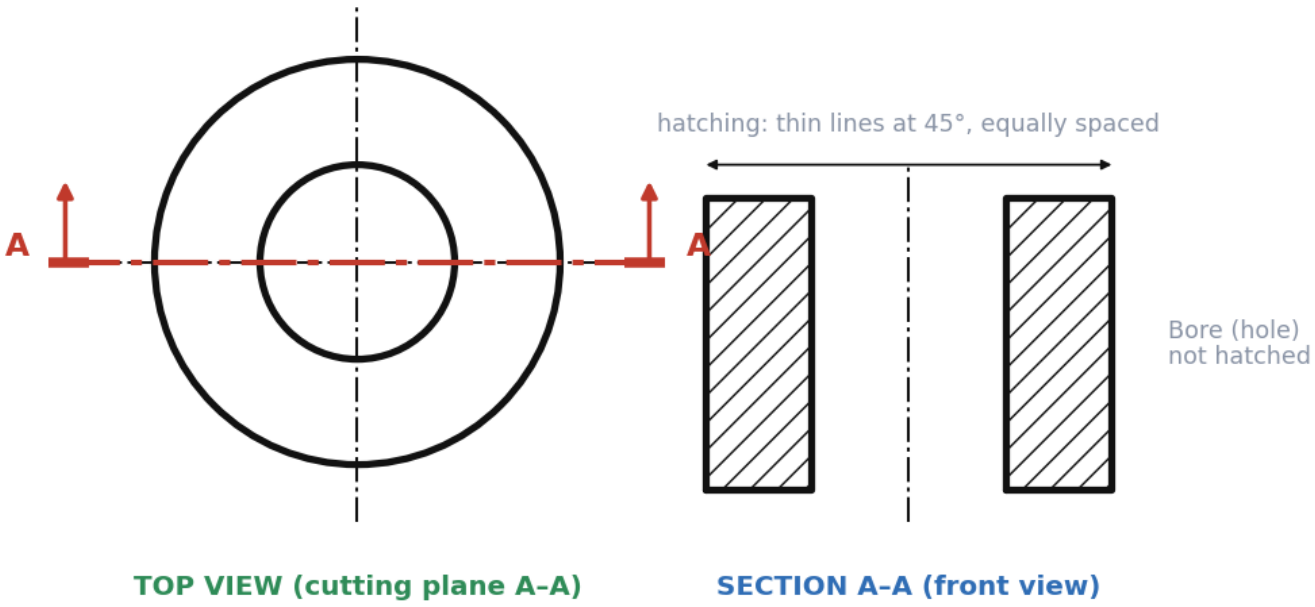



*Fig. 3.1 — Sectional view of a bush: cutting plane A–A and hatched section*

- **Cutting plane line**: chain thin, thick at ends and where direction changes; arrows show the direction of viewing; label A–A.
- **Hatching**: thin continuous lines at **45°**, equally spaced (1.5 – 3 mm). The same direction & spacing for one part; opposite direction for adjacent parts.
- Holes, voids and the **space** where nothing is cut are **not hatched**; hidden lines are omitted in the sectional view unless needed.
- Ribs, webs, shafts, bolts, keys are **not sectioned** when the cutting plane passes longitudinally through them.

**APPLICATIONS**

Sectional views reveal **internal features** of engine blocks, valves, pumps, bearings and pressure vessels – without cluttering the drawing with hidden lines.

## 4. Development of Surfaces

**DEFINITION**

**Development** is the unfolding of the lateral (and base) surfaces of a solid on a plane, so that every face appears in its **true shape and size**, laid flat in one piece. It is the reverse of folding a sheet into a solid.

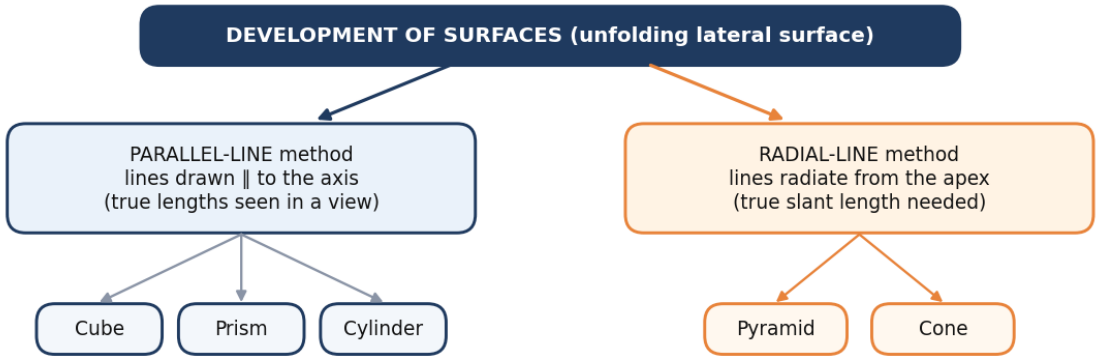



*Fig. 4.1 — Methods of development*

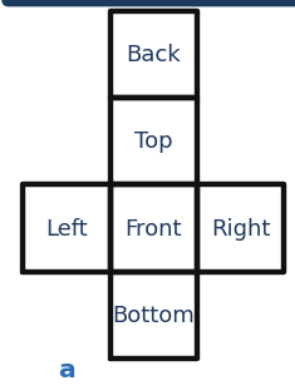
**FORMULAS**

Lateral surface area (prism) = **perimeter of base × height** | Cylinder: **πDh** | Pyramid: **½ × perimeter × slant height** | Cone: **π r l**

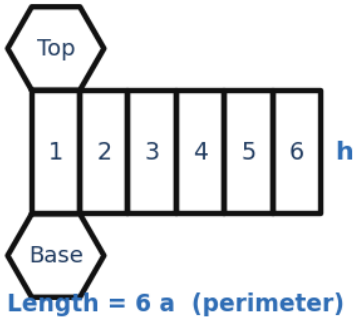
Slant edge of regular pyramid: **l = √(h² + R²)**, R = circumradius of base (square: a/√2) | Cone: **l = √(r² + h²)**, sector angle **θ = (r / l) × 360°**

## 4.1 Parallel-line development

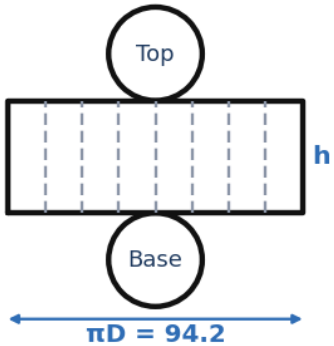
**(a) Cube – 6 squares (cross shape)**




**(b) Hexagonal prism – rectangle 6a × h**




**(c) Cylinder – rectangle πD × h**




**(d) Quick reference**

| Solid | Development | Lateral area |
|---|---|---|
| Cube | 6 squares | 4 a² |
| Prism | rectangle P × h | P × h |
| Cylinder | rectangle πD × h | π D h |
| Pyramid | sectors of triangles | ½ × P × l |
| Cone | sector, angle = (r/l)·360° | π r l |

*Fig. 4.2 — Development of cube, hexagonal prism and cylinder*

**PROCEDURE**

1. Draw the stretch-out line equal to the **perimeter** (cube 4a, prism n·a, cylinder πD) and mark the divisions of the base edges / 12 equal parts of the circle.
2. Erect **perpendiculars of height h** at each division; join the tops – this is the lateral surface.
3. Attach the **top and base** (true shape) to one edge of the rectangle; for cylinders attach them tangentially at the middle.
4. Mark fold lines as thin dashed lines and show edge numbers.

## 4.2 Radial-line development

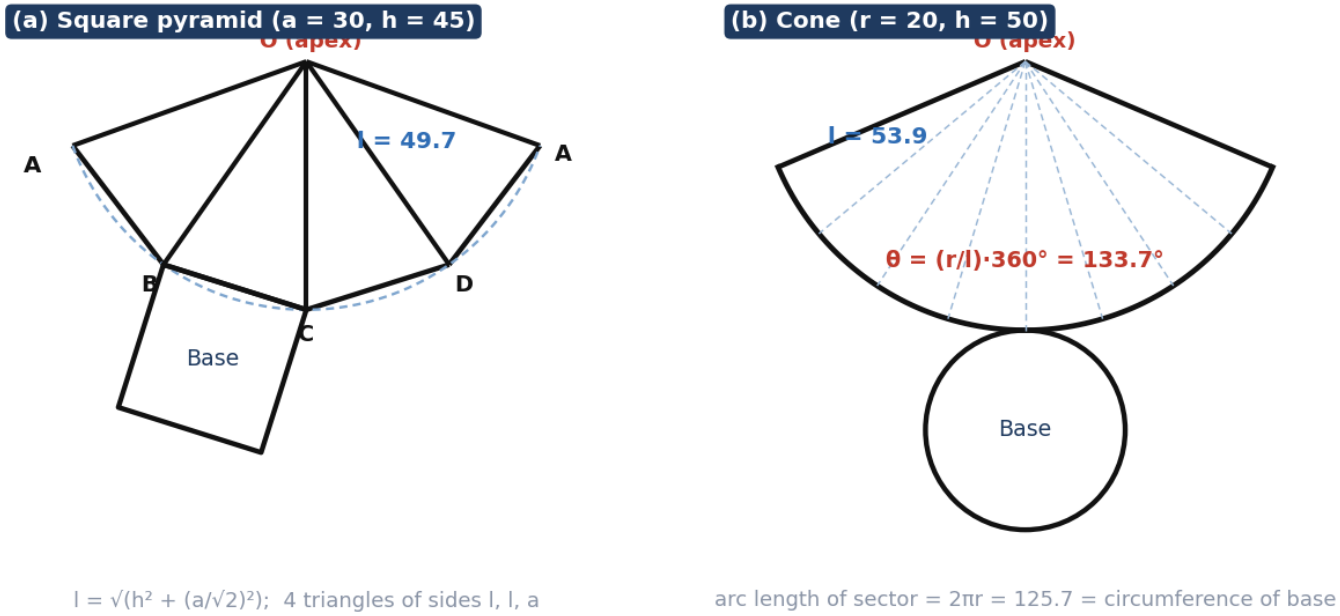



*Fig. 4.3 — Development of a square pyramid and a cone*

**PROCEDURE**

1. Find the **true slant length l** (draw the FV of the slant edge ∥ to VP or calculate it).
2. **Pyramid:** with apex O and radius l draw an arc; step off the **base-edge length a** chords **n times**; join to O. Attach the true-shape base to one edge.
3. **Cone:** with radius l draw a sector of angle **θ = (r/l)·360°**; its arc length equals the base circumference 2πr. Attach the base circle tangentially.
4. For a **frustum**, subtract the smaller sector (radius l′ = l × $r_2/r_1$) from the larger one.

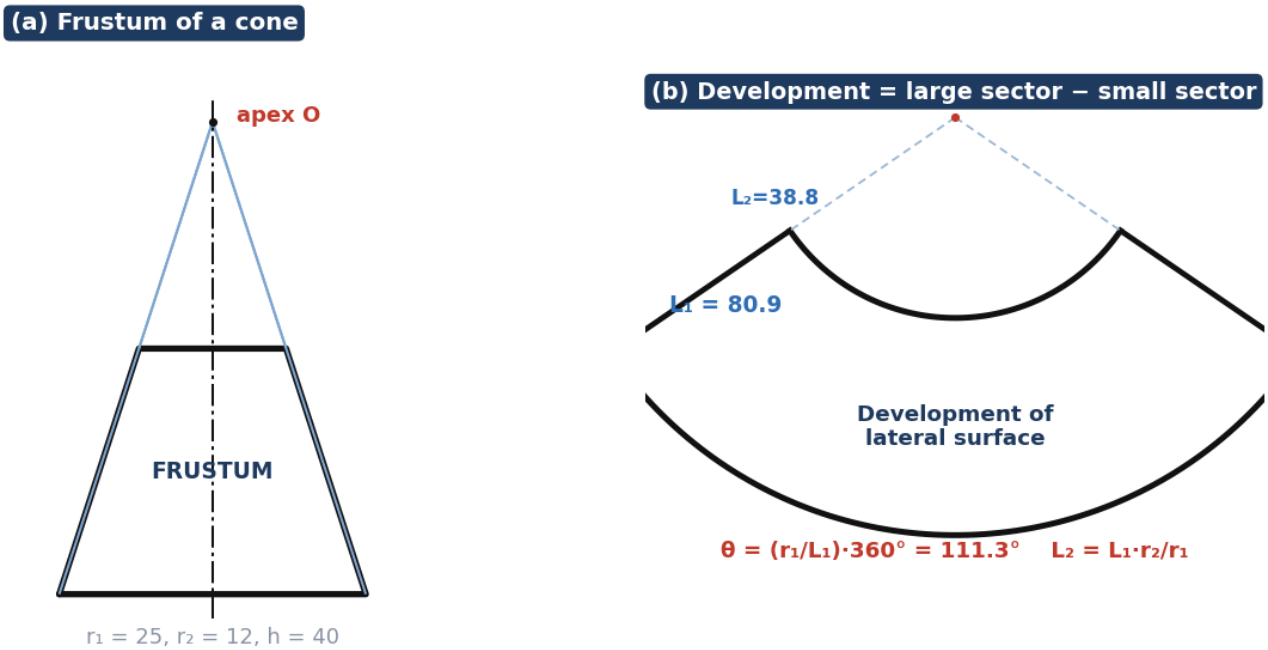



*Fig. 4.4 — Development of a frustum (truncated cone)*

**SOLVED EXAMPLE**

**Development of a cone, base diameter 40 mm, axis 50 mm.**

r = 20; l = √(20² + 50²) = **53.9 mm**; θ = (20 / 53.9) × 360° = **133.6°**. Draw a sector of radius 53.9 mm and angle 133.6°; arc length = 2π × 20 = 125.7 mm ✓.

**APPLICATIONS**

Development is the basis of **sheet-metal work**: ducts, funnels, hoppers, chimneys, tanks, packaging cartons and pattern-making for fabrication.

## 5. Quick Revision & Exam Questions

**QUICK SUMMARY**

- Section ∥ HP → true shape in TV; section ∥ VP → true shape in FV; otherwise use an **auxiliary view**.
- Hatch the cut face at **45°**; cutting plane line = chain thin, thick ends.
- Plane inclined to a cone's axis → **ellipse**; ∥ generator → parabola; steeper → hyperbola.
- Parallel-line: prisms & cylinders (rectangle). Radial-line: pyramids & cones (sectors).
- Cone sector angle θ = (r / l) × 360°; cylinder rectangle = πD × h.

**EXAM-ORIENTED QUESTIONS**

**Q1.** Define section plane, sectional view and true shape of a section. **[3]**

**Q2.** A square prism (side 40, height 70) has its axis ⊥ HP. A plane ⊥ VP and inclined 45° to HP cuts the axis at 35 mm from the top. Draw the FV, sectional TV and true shape. **[8]**

**Q3.** A hexagonal pyramid (side 30, axis 65) is cut by a plane ⊥ VP inclined 40° to HP bisecting the axis. Draw FV, sectional TV and true shape. **[8]**

**Q4.** A cone (base Ø50, axis 70) is cut by a plane inclined 45° to HP, passing through the axis midpoint. Draw the sectional views and true shape. **[8]**

**Q5.** Draw the development of a cube of side 35 mm. **[6]**

**Q6.** Develop the lateral surface of a hexagonal prism (side 25, height 50). **[6]**

**Q7.** Develop a cylinder Ø40 × 60; show the development of a point 20 mm from the base on a generator. **[8]**

**Q8.** Develop a square pyramid, base 40 mm, axis 60 mm. **[8]**

**Q9.** Develop a cone of base Ø50 and axis 70 mm. Find the sector angle. **[8]**